

SCADA-Anchored Regime-Aware Routing for Auditable Wind-Turbine Control-Boundary Forecasting

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Abstract—Wind-farm reserve screening near the MPPT-to-pitch transition depends on knowing which control law is active, yet the threshold label that identifies it can be delayed, missing, or noisy exactly where over-forecasts create shortage exposure. We make this control-boundary assignment auditable through SCADA-anchored, regime-aware routing. A node-level mixture-of-experts gate is constrained by operating anchors so each turbine-time route can be checked against a declared MPPT-to-pitch partition and issued before future active power is observed. On the KDD Cup 2022 benchmark, recomputing alignment and forcing class weights from the training split only keeps the routed forecaster below the RMSE guardrail (229.93 versus 224.34 for iTransformer) while preserving the routing claim (NMI 0.721, ARI 0.740); the fully archived legacy checkpoint gives stronger route alignment (NMI 0.87, ARI 0.92) for the full operational audit. Under a six-step label delay, the route retains 0.960 early pitch-window recall while the delayed threshold rule falls to 0.196; under 50% label availability, the corresponding rule reaches 0.508. A simple issue-time anchor classifier reaches 1.000 recall and 0.879 precision on clean anchors, so the claim is not superior standalone detection: the contribution is an auditable in-model route assignment coupled to forecasting and boundary-window reserve screening. At shortage-to-reserve cost ratio 10, gate-conditioned binning lowers same-router boundary-window shortage energy by 14.5% and violation by 1.38 percentage points while carrying about 3.6% more reserve; seed-paired intervals support this same-router diagnostic, but physical-bin baselines remain competitive. Retrained on the independent EN-GIE La Haute Borne farm, the router provides anchor-observable cross-site mechanism replication (five-seed NMI 0.941); Kel-marsh/Penmanshiel, lacking pitch observability and boundary support, define the deployment conditions required before cross-farm use. The contribution is an auditable control-boundary routing layer with degraded-label availability evidence, a bounded RMSE guardrail, same-router reserve-screening evidence, and explicit transfer conditions.

Index Terms—Wind-turbine control boundary; SCADA-anchored routing; regime-aware forecasting; auditable routing; MPPT-to-pitch transition; mixture-of-experts.

I. INTRODUCTION

Wind-farm reserve screening near the transition from maximum power point tracking (MPPT) to blade-pitch control depends on knowing which control law is currently active, yet the threshold label that identifies it can arrive late, go missing, or turn noisy exactly where over-forecasts create shortage exposure. This makes the transition both operationally decisive and poorly observed: a forecaster must answer not only how large the error is, but which operating state generated the

local response. In the MPPT region, power responds strongly to wind-speed variation; as pitch control activates near rated operation, the local power-response law changes. A small forecast error near rated wind can cross a different control law, yet the transition occupies a smaller share of normal operation, so its effect can be diluted in fleet-average RMSE while still concentrating local risk. A useful forecasting system for this setting must therefore expose the operating boundary that produced each forecast and stay usable when the label stream degrades, not only report aggregate accuracy.

State-of-the-art spatio-temporal forecasters now model turbine coupling, wake interaction, dynamic dependence, and sensor-rich wind-farm layouts [1], [2], [3], [4], [5]. These models set the accuracy reference, but a low-error graph encoder does not by itself tell a reserve planner whether the current local map is MPPT-like or pitch-control-like. Mixture-of-experts (MoE) routing can separate heterogeneous response laws, yet a gate trained only through prediction loss can specialize on partitions that have no operating meaning [6], [7], [8], [9], [10]. The missing middle layer is an auditable routing assignment that identifies which physical response law is active at the anchor time and can then be used in reserve-risk diagnosis.

We make the routing decision itself the operating-boundary diagnostic. A node-level MoE gate is constrained by SCADA operating anchors so that each turbine-time assignment can be compared with a declared MPPT-to-pitch partition and issued before future active-power outcomes are observed. This supports degraded-label reserve screening: when thresholds are late, incomplete, or noisy, the operator still has a signal for early pitch-control entry. WTB is the source control-boundary benchmark because the aerodynamic boundary is partly hidden inside turbine-control action; ERA5 is retained as an observability contrast where the thermodynamic marker is more directly visible through sensible heat flux. The recovered gate is connected to validation-calibrated, test-frozen reserve diagnostics that report cost, violation rate, reserve energy, and shortage energy under declared shortage-to-reserve cost ratios. We further compare gate-bin reserve allocation with validation-frozen global and physical-bin quantile baselines. This positioning deliberately separates transition-window accountability from the conventional forecasting leaderboard and from claims of full dispatch optimality.

The operator-facing workflow is deliberately narrow. At issue time the model may read the current SCADA anchors, historical active power, graph state, and missingness masks; it may not read future active power. A confirmed regime label, in contrast, may be delayed by validation, telemetry drop-out, or quality-control logic. The router is therefore not proposed as an anchor-free discovery device or as a better standalone classifier than a simple rule or logistic model on the same live anchors. Its role is to make the boundary assignment part of the forecasting model, so the same saved run can be audited for route semantics and then sliced for reserve-risk diagnosis.

This paper makes three contributions. First, it introduces SCADA-anchored regime-aware routing as an auditable MPPT-to-pitch assignment whose semantics are fixed by operating anchors rather than post-hoc expert interpretation. Second, it quantifies degraded-label availability while disclosing a simple issue-time classifier control: the routed gate keeps early pitch-window recall at 0.960 under six-step label delay and 50% label availability, threshold-label rules fall to 0.196 and 0.508, and a logistic anchor classifier reaches 1.000 clean-anchor recall, showing that the detection value comes from issue-time anchor availability rather than a uniquely superior gate classifier. Third, it delivers this operating-state layer with an explicit accuracy guardrail and bounded operating consequence: a training-split class-weight rerun preserves the route at RMSE 229.93 (5.59 units above iTransformer), the archived full-provenance route supplies same-router reserve-screening evidence with seed-paired uncertainty support, and ENGIE La Haute Borne provides anchor-observable cross-site mechanism replication at five-seed routing NMI 0.941. The Kelmarsh/Penmanshiel coverage gaps are converted into explicit deployment conditions rather than claims of automatic cross-farm generalization. The framing throughout is a complementary diagnostic layer for an already-deployed low-RMSE forecaster, not a replacement on the forecasting leaderboard or a standalone operating-state classifier.

II. RELATED WORK

A. From average wind-power accuracy to transition-window risk

Wind-power forecasting has moved from single-site prediction toward spatio-temporal graph models that encode physical sources of error. Reviews identify ramp and uncertainty as recurrent operational risks [11], [12], while graph mechanisms provide the accuracy reference [1], [13], [14]. Wind-specific extensions add wake-aware graphs, SCADA fields, and physics-guided constraints [2], [4], [5]. The remaining gap is diagnostic: these models set the low-error benchmark, but they do not make the active turbine-control law visible to a reserve planner.

Regime-aware MoE routing separates local response laws, yet prediction loss alone need not produce a physically meaningful partition [8], [9], [10]. Physics-guided learning offers the missing constraint principle [15], [16], [17], but for wind turbines that principle must pass through SCADA observability: operating-state labels depend on wind speed, pitch, and active

power [18], [19]. The distinction we draw is where the physics enters: prior MoE and physics-guided forecasters constrain the prediction output or let prediction loss organize the experts, whereas we constrain the routing decision itself against a declared operating boundary. The gate assignment, not only the forecast, is therefore the audited object, and downstream reserve logic can consume it as a physically traceable signal.

Forecast value is realized through reserve and commitment decisions. Reserve studies show that variable generation changes requirements [20], [21], while quantile methods bridge point forecasts to decision risk [22], [23]. The diagnostic question here is earlier in the chain: which operating window concentrates reserve exposure, and can the forecast model reveal that window through a physically auditable route assignment? The proposed reserve audit is complementary to probabilistic dispatch and market-clearing models.

III. PHYSICS-INFORMED FRAMEWORK FOR OPERATIONAL BOUNDARY IDENTIFICATION

A. Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. Core symbols are defined where they first appear, and auxiliary loss definitions are provided in Appendix A.

B. Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularization. This engineering layer defines what counts as a control regime and when that regime is observable from the data.

1) *WTB operating regimes*: In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}).$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. The regime labels use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , and writing $w_{i,t} = \text{Wspd}_{i,t}$ for compactness, we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

The reported implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$ (Appendix A). These thresholds are declared operating anchors, fixed before test evaluation and then audited by threshold-grid sensitivity checks; they are not universal turbine constants and are not selected by test-set NMI.

2) *ERA5 observability contrast (see Supplementary Material)*: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the WTB control-boundary accountability task.

C. Shared architecture

1) *Physical graph construction*: The graph is part of the physical problem specification. WTB uses a directed wake graph: candidate turbine pairs are filtered by proximity, activated when an upstream turbine lies inside a wind-aligned downstream cone, weighted by streamwise and cross-stream decay, and pruned to the strongest inbound neighbors. The same directed weights define a wake score, which supplies the WTB wake auxiliary label on MPPT and pitch-control samples. ERA5 uses a symmetric Haversine-Gaussian k_{nn} graph because the thermodynamic regime marker is already visible in the observed state. Appendix A gives the graph equations and constants.

2) *Directed-diffusion GRU encoder and node-level gate*: The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = \text{LN} & \left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) \right. \\ & \left. + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)} \right), \end{aligned} \quad (1)$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is standardized from training-split statistics and is never filled from the prediction horizon.

3) *Routing stack and evidence boundary*: This makes the routing stack an anchor-constrained diagnostic rather than an unsupervised operating-state discovery method. High gate-regime agreement means that the implemented router obeys the declared SCADA boundary under the available anchor set; it does not establish anchor-free regime recovery.

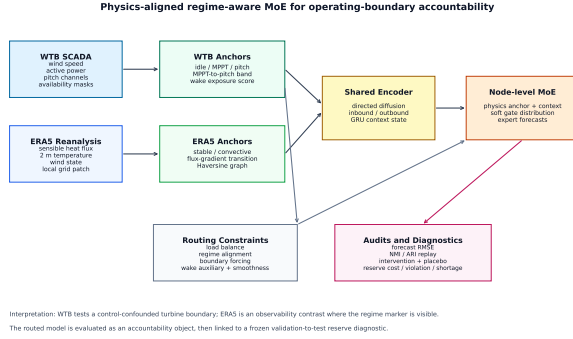


Fig. 1. Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

D. Integration of Physical Constraints into Model Optimization

Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [8], [9]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (2)$$

with inactive terms set to zero. Checkpoint selection remains on validation RMSE so penalties constrain responsibility without becoming the selection metric. Exact weights are in Supplementary Appendix~A.

1) *Load balancing*: A balancing term prevents early expert collapse before regime-specific structure has time to emerge. It couples hard top- K usage with soft probability mass over valid node-time samples. Supplementary Appendix~A gives the exact expression; physical ownership is supplied only by the anchored alignment terms below.

2) *Regime-anchor alignment*: The alignment term connects the first C gate logits to the declared regime structure so that cross-seed statements such as “MPPT-aligned” refer to a fixed mapping rather than to post-hoc relabeling. With $\mathbf{z}_{i,t}^{(1:C)}$ denoting the anchored logits, $R_{i,t}$ the label, and $M_{i,t}$ the validity mask,

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}),$$

$$\Omega = \{(i,t) : M_{i,t} = 1\}.$$

with inverse-frequency class weights. The released preprocessing code computes these weights on the training split; the evidence package includes a class-weight provenance audit and train-only sensitivity rerun for the frozen strict-cache

checkpoints. In that five-seed rerun, the boundary-forced router keeps strong route recovery (NMI 0.721, ARI 0.740) while keeping the same strict-mask evaluation boundary.

3) *Boundary-focused forcing (WTB only)*: After coarse alignment the boundary remains partly masked by control action. A focused forcing term acts on MPPT and pitch-control samples only. With $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ for $R_{i,t}^{\text{wtb}} \in \{1, 2\}$,

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}([z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}]^T, Y_{i,t}^{\text{force}}),$$

where $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$. This term is not a lead-time switch predictor; it is a boundary identifiability regularizer for the issue-time operating state.

4) *Wake auxiliary supervision and graph smoothness*: A wake auxiliary label identifies residual wake ambiguity inside MPPT and pitch-control samples, and a graph-smoothness penalty discourages noisy neighbor-to-neighbor gate jumps. Both formulas are in Supplementary Appendix~A.

E. Operational reserve diagnostic protocol

The reserve diagnostic tests whether a physically auditable gate changes the reserve tradeoff in the MPPT-to-pitch window after the point-forecast model has been trained. For each saved WTB run, validation predictions select reserve levels; test predictions are then evaluated once with those levels fixed. The protocol follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ the realized active power at an evaluated horizon cell. We define over-forecast shortfall as

$$s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0),$$

because this is the error direction that leaves a reserve schedule exposed. For a reserve policy b and empirical quantile q , the validation split estimates a reserve level

$$r_{b,q} = Q_q(\{s_{i,t} : (i,t) \in b, (i,t) \in \mathcal{V}\}),$$

where Q_q is an empirical quantile and $\mathcal{V}$ is the validation split. The grid is fixed before test evaluation:

$$q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}.$$

The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{V}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{V}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t.$$

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. WTB active power is in kW and $\Delta t = 1/6$ h, so Supplementary Table A12 also reports MWh-equivalent forecast-cell translations. These are not delivered market energy or settlement costs: a value written as 84.58M denotes 84.58 million reserve-cost-equivalent kWh cells, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ charges one unit of residual shortage as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware screening where shortage avoidance dominates local cost savings. The primary diagnostic compares Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin policies. Only same-model global comparisons attribute gate-bin reserve effects to the learned router.

The market-facing use case is therefore a pre-dispatch reserve and imbalance-risk screening log rather than a settlement model: the saved run records route probability, reserve bin, selected quantile, residual shortfall, and anchor state for each escalated transition-window cell. A modular low-RMSE forecaster plus issue-time classifier can also emit a state flag; the extra value claimed here is narrower-same-run responsibility attached to the forecast residual and reserve slice, not superior clean-anchor classification or market-clearing profit.

To bound this diagnostic against a probabilistic reserve alternative, we additionally run validation-frozen empirical quantile baselines for the global and physical operating bins. These are not trained distributional forecasters; they are conformal-style shortfall reserves estimated on the validation split and evaluated once on the test split. The main reported boundary window is fixed at ± 1.0 m s⁻¹ around rated wind and the main cost ratio is $\rho = 10$, with sensitivity at $\rho \in \{2, 5, 10, 20, 50\}$. A separate toy operational-cost module reports reserve procurement plus ρ -weighted residual-shortage penalty and excludes optimal power flow, unit commitment, market clearing, delivery constraints, and price claims.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [19]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then

completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [24]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

B. Techno-Economic Validation and Benchmarking Framework

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE are included for WTB [25], [26], [27], together with deterministic persistence, a fitted physical power-curve baseline, XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split avoids using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**. The main WTB forecasting table reports representative strong baselines, including the best strict-cache baseline used for the RMSE price, while the full neural, anchor-only, and engineering baseline records remain in the source tables. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Supplementary Appendix~A so that the Results section can focus on evidence rather than model bookkeeping.

C. Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on val-

idation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in Supplementary Appendix~A so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205, giving 30 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

V. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The results follow the operating decision that motivates the paper: can a route replace a degraded threshold stream in the MPPT-to-pitch window, and what accuracy trade-off does that issue-time signal carry? We first test boundary recovery and falsification checks, then quantify early pitch-window recall under delayed, missing, and noisy labels. We then ask whether the audited gate changes a validation-frozen reserve rule, and finally state the deployment gates for new wind farms.

A. Can the Gate Recover the Operating Boundary?

SCADA-anchored routing recovers the WTB MPPT-to-pitch partition strongly enough to make the gate an auditable operating-boundary signal. Across five WTB seeds, the training-split class-weight rerun remains above the routing-claim threshold at NMI 0.721 and ARI 0.740 while reaching overall RMSE 229.93 $\pm$ 2.50. This narrows the accuracy gap to 5.59 units relative to iTransformer (224.34 $\pm$ 2.23) and 4.19 units relative to Graph WaveNet (225.74 $\pm$ 2.60). The fully archived legacy strict-cache boundary-forced router has a larger accuracy price (236.13 $\pm$ 8.41) but stronger

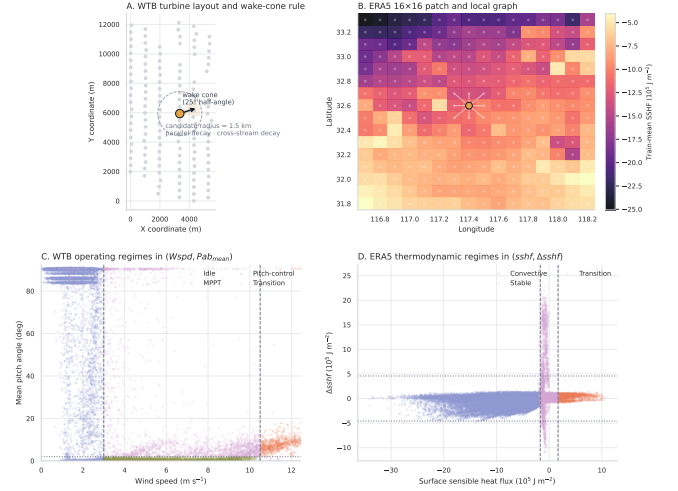


Fig. 2. Operating-decision context and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split, included as an observability contrast.

alignment (NMI about 0.87, ARI about 0.92) and supplies the saved gate/prediction artifacts used for the downstream label-degradation and reserve audits. The intended operating point is therefore not leaderboard replacement, but an additional issue-time state signal for the transition window with its accuracy price made explicit.

The boundary itself is a meaningful diagnostic target. Anchors within $\pm 1.0 \text{ m s}^{-1}$ of rated wind have RMSE 321.17 $\pm$ 21.31, roughly 67 units above valid non-boundary MPPT/pitch anchors. The same boundary band has mixed-regime structure (NMI/ARI 0.6562/0.7766), so it is both a forecasting stress point and a natural reserve-risk window. ERA5 provides the positive observability contrast: when the regime marker is directly visible through sensible heat flux, routing correction improves alignment without changing the headline accuracy story.

TABLE I
OPERATING-BOUNDARY RECOVERY AND MECHANISM CHECKS.

Check	Metric	Result
WTB boundary-forced router	Gate-regime NMI / ARI	0.8716 +/- 0.0418 / 0.9166 +/- 0.0371
Train-only class-weight rerun	Gate-regime NMI / ARI; overall RMSE	0.7208 / 0.7398; 229.93
WTB boundary-band audit	Boundary-band RMSE; mixed-regime NMI / ARI	321.17 +/- 21.31; 0.6562 / 0.7766
Boundary-anchor intervention	Overall RMSE change; NMI / ARI drop	+2.6792; 0.7017 / 0.8593
WTB spatial and time holdouts	Held-out-node NMI / ARI; future-holdout NMI / ARI	0.8340 / 0.8829; 0.8352 / 0.8870
La Haute Borne cross-site replication	NMI / ARI; Patv-zero NMI; wind+pitch-zero NMI	0.9408 +/- 0.0336 / 0.9712 +/- 0.0201; 0.953; 0.001
Threshold-grid audit	Worst-case saved-gate NMI / ARI	0.8655 +/- 0.0429 / 0.9146 +/- 0.0377

The diagnostic survives the main falsification checks. Boundary-anchor zeroing collapses route agreement, wake-score zeroing is near-null, and placebo labels do not reproduce the actual-label agreement. Within WTB, the gate recovers the declared boundary on withheld turbines (NMI 0.834) and a held-out future period (NMI 0.835). On the independent ENGIE La Haute Borne farm, retraining from scratch reaches five-seed NMI 0.941 and ARI 0.971 with directly observed pitch; zeroing active power still gives NMI 0.953, whereas jointly zeroing wind speed and pitch collapses NMI to 0.001 (Supplementary Table A9). Thus La Haute Borne is anchor-observable cross-site mechanism replication, while Kelmarsh/Penmanshiel remain deployment-gate failures caused by insufficient pitch observability and boundary-cell support.

The central operating result is an availability audit rather than a claim that the gate is the best standalone detector. We audit saved five-seed gates around MPPT-to-pitch transitions after delaying threshold labels, dropping label availability, and recomputing labels under sensor noise. Table-II reports recall on early pitch-control cells, and Supplementary Table A10 adds precision plus a simple issue-time logistic classifier control. With a six-step delay, the threshold rule recalls only 0.196 of early pitch cells, whereas the gate recalls 0.960 and recovers about 566 early pitch-window cells per seed. With 50% label availability, the rule recalls 0.508 while the gate remains at 0.960. On clean issue-time anchors, however, the logistic classifier reaches 1.000 recall and 0.879 precision, above the gate precision of 0.759. The result therefore shows that live SCADA anchors preserve early-window state information when the confirmed label stream degrades; the MoE contribution is that this state assignment is embedded in the forecaster and can be audited and sliced, not that it dominates all issue-time classifiers.

This audit asks about a degraded confirmation stream, not clean-anchor classifier ranking. A modular low-RMSE forecaster plus live-anchor classifier and physical-bin reserve rule is a valid alternative when only a state flag is needed. Its missing piece is trainable route responsibility attached to

each forecast and same-run route-conditioned residual/reserve slicing. A direct Graph WaveNet+classifier-bin reserve control improves over Graph WaveNet/global (86.54M versus 88.80M) but trails physical-bin and boundary-gate references (84.31M and 84.58M; source table in the evidence package), so it is a tested modular control rather than a replacement for in-model responsibility. The route-evolution audit adds the dynamic check: gate matching to the new regime peaks at the transition step (0.815) and drops under three- and six-step lead shifts (0.361 and 0.405; Supplementary Table A10b). The anchor-only router has lower overall RMSE (233.46) and slightly higher NMI/ARI (0.8895/0.9219), but worse pitch-control RMSE (317.81 versus 286.95) and far larger 0.5-std boundary-anchor-noise degradation (105.56 versus 5.78).

We report the operating trade-off explicitly. A model receives citable degraded-label gain only after its route passes the physical-routing audit; otherwise the accountability gain is set to zero, even for an accurate or routed-capacity model. Under this rule iTransformer is the zero-RMSE-price anchor without an audited operating-state route, Unconstrained MoE fails the route audit (NMI 0.014), and the fully archived boundary-forced checkpoint is the complete operational-audit run that carries a larger historical RMSE cost for an audited issue-time route and +0.764 six-step-delay recall gain. The train-only class-weight rerun shows that the same routed operating layer can be kept within a 5.59-RMSE gap, while the legacy checkpoint remains the run with complete gate, prediction, and reserve artifacts. The logistic control is reported separately because it is a detector on live anchors, not a forecaster with routed attribution.

TABLE II
LABEL-DEGRADATION AUDIT FOR EARLY MPPT-TO-PITCH PITCH-WINDOW DETECTION.

Scenario	Setting	Gate	Rule	Gain
Delay	1 step	0.960	0.655	+0.305
Delay	3 steps	0.960	0.381	+0.579
Delay	6 steps	0.960	0.196	+0.764
Availability	75% labels	0.960	0.744	+0.216
Availability	50% labels	0.960	0.508	+0.452
Availability	25% labels	0.960	0.242	+0.718
Noise	Wspd 0.5, Pab 1.0	0.960	0.808	+0.152
Noise	Wspd 1.0, Pab 2.0	0.960	0.652	+0.308

The training-level anchor-stress guard also keeps the leakage boundary visible. Across five seeds and 20 completed variant runs, removing Patv or Pab_mean leaves mean NMI at 0.881 and 0.878, while lagging Patv and lagging pitch/wind give 0.763 and 0.684. We therefore use “partial anchor robustness” wording: the route is not a leakage artifact, but it is still a declared-anchor mechanism rather than anchor-free discovery.

B. Boundary-Risk Vignette: Does the Gate Change Reserve Tradeoffs?

The reserve audit is a consequence check for the recovered route, not a dispatch optimizer. A validation-calibrated reserve screener is frozen and applied to later MPPT-to-pitch boundary

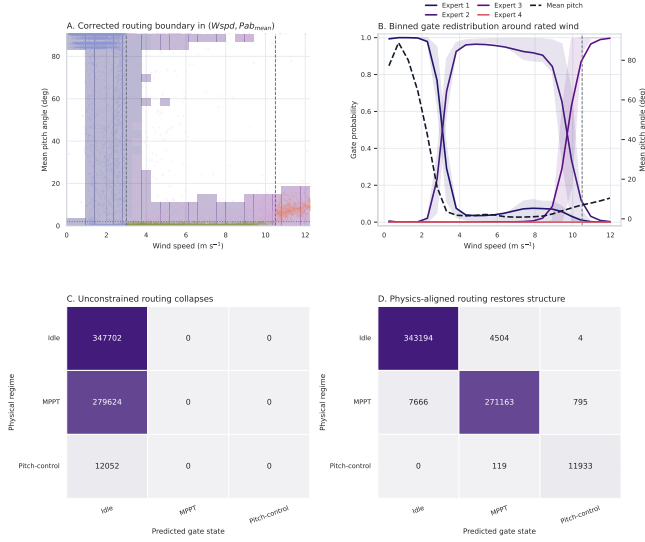


Fig. 3. The WTB operating plane shows the main mechanism: after correction, the dominant routed responsibility changes around the rated-wind and pitch-control boundary instead of forming an arbitrary expert partition. The confusion matrices summarize the same recovery numerically.

cells, comparing global, physical-bin, and gate-bin quantile rules for Graph WaveNet and the boundary router.

At shortage-to-reserve cost ratio 10, gate-conditioned binning carries about 3.6% more reserve for 14.5% lower shortage energy and a 1.38 percentage-point lower violation rate than the same routed model with a global rule. Seed-paired intervals for this same-router boundary comparison stay below zero for total cost, violation, and shortage (Supplementary Table A11). Physical-bin baselines remain competitive and cross-backbone/full-sample uncertainty remains unfavorable, so the claim is a statistically supported same-model transition-window diagnostic, not reserve-policy superiority.

TABLE III
BOUNDARY-WINDOW RESERVE DIAGNOSTIC AND VALIDATION-FROZEN QUANTILE BASELINES AT COST RATIO 10. SAME-MODEL GATE-BIN DIFFERENCES HAVE SEED-PAIRED SUPPORT; PHYSICAL-BIN AND FULL-SAMPLE COMPARISONS BOUND THE CLAIM.

Policy	Cost	Viol.	Reserve	Shortage	Reading
Boundary/physical-bin	83.78M	0.0880	53.26M	3.05M	Best same-router physical-bin reference
GWN/physical-bin	84.31M	0.0931	50.35M	3.40M	Low-RMSE backbone reference
Boundary/gate-bin	84.58M	0.0900	52.67M	3.19M	Gate diagnostic; same-model better than global
Boundary/global	88.13M	0.1038	50.82M	3.73M	Same-model global rule
GWN/global	88.80M	0.1022	50.32M	3.85M	Global low-RMSE reference

The useful reserve window is moderate rather than universal: ratios 5–10 favor gate-bin screening, ratio 50 favors the same-model global rule, and Graph WaveNet/physical-bin remains close at 84.31M. The boundary-slice proxy falls from 88.13M to 84.58M, which Supplementary Table A12 translates into

about 540 avoided MWh-equivalent shortage cells and a 355k illustrative reserve-cost-scale marker at 100 EUR/MWh. The paper therefore claims pre-dispatch screening value, not market revenue or system-wide dispatch value.

C. What Are the Costs and Deployment Gates?

The accuracy cost is real. iTransformer, Graph WaveNet, and lag-feature baselines remain better whole-sample forecasters on WTB; the routed model is useful when an operator needs an accountable operating-state assignment for a boundary-specific decision. Time-forward testing similarly shows that stable routing semantics do not guarantee stable value prediction under distribution shift.

The external evidence is deliberately two-sided. La Haute Borne passes the anchor-observable replication check (NMI 0.941, ARI 0.971, with replay support in Supplementary Tables A8–A9). Kelmarsh/Penmanshiel do not authorize gate-bin reserve use because pitch observability and boundary coverage fail. Before a new farm uses the route, it must verify pitch or proxy observability, boundary-cell support, compatible geometry, local threshold estimation, and a held-out routing pass.

TABLE IV
EXTERNAL-SITE DEPLOYMENT GATES IDENTIFIED BY KELMARSH/PENMANSHIEL TESTING.

Gate	Observed evidence	Required decision before reserve use
Held-out routing criterion	Default WTB boundary gives mean external NMI 0.4877 and ARI 0.5112, below the 0.50 NMI criterion	Re-establish routing agreement locally
Local boundary recalibration	Average test NMI changes from 0.1324 to 0.1491 after validation-only recalibration	Treat threshold transfer as insufficient
Pitch/proxy observability	One transfer direction has no effective pitch-feature coverage or boundary cells	Require pitch channels or a validated proxy
Calibration-window adaptation	Small-window adaptation remains below the held-out balanced-accuracy threshold	Use calibration only after a held-out pass
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation

VI. LIMITATIONS

The WTB correction uses threshold pseudo-labels from wind speed and mean pitch angle, and those same channels appear in the gate anchor. The method therefore constrains a router to a declared operating boundary rather than discovering an anchor-free physical state. The active-power anchor is also a timing constraint: inputs may contain $\text{Patv}_{t-H+1:t}$ and issue-time Patv_t , while targets begin at $\text{Patv}_{t+1:t+P}$.

The expert-regime language is tied to the implemented logit convention: anchored logits are interpretable under the declared mapping, while unassigned experts and wake auxiliary logits should not be overread as universal turbine states. The

comparison is bounded to the implemented WTB baselines and validation-frozen quantile reserve checks; trained quantile forecasters, scenario reserve, OPF, unit commitment, market clearing, and security-constrained reserve are outside scope.

The strongest claims are five-seed WTB routing recovery, mechanism intervention, within-WTB spatial/temporal stress, La Haute Borne anchor-observable replication, and same-router boundary reserve diagnostics. Gate-alignment deltas versus the full MoE and cross-backbone reserve comparisons remain bounded diagnostics; automatic cross-farm reserve use is outside authorized claims unless observability and held-out routing gates pass.

VII. CONCLUSION

SCADA-anchored regime-aware routing turns the node-level gate into an auditable MPPT-to-pitch assignment inside the forecaster. The training-split class-weight rerun preserves the routing claim at NMI 0.721 and ARI 0.740 while keeping overall RMSE at 229.93, within 5.59 units of the best strict-cache baseline; the fully archived legacy checkpoint recovers the declared WTB partition more strongly (NMI about 0.87, ARI about 0.92) and supplies the complete operational audit artifacts. The route keeps 0.960 early pitch-window recall under a six-step label delay while the delayed threshold rule falls to 0.196 and exposes transition-window reserve risk. A simple issue-time anchor classifier is disclosed as a stronger clean-anchor detector, so the claim remains in-model route attribution rather than classifier superiority. Same-router gate-bin reserve screening has boundary-window statistical support, while validation-frozen physical-bin baselines and deployment gates keep the claim bounded.

VIII. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

IX. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmarsh, and Penmanshiel SCADA datasets are public; raw third-party data are not redistributed. Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress cache/train/guard, early-warning label degradation, classifier control, class-weight provenance/sensitivity, external diagnostics, and evidence-freeze protocols. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

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